Strategic Resource Allocation across Operator Platforms: A Game-Theoretic Auction Approach

Appendix

The CAMARA-compliant *VERA* API endpoints given in Table I are designed with explicit role separation between the *Federated Resource Arbiter (FRA)* and participating *MECs*, ensuring controlled and auditable federation-wide operations. The `POST /mecs` endpoint is invoked solely by the FRA to register a MEC by recording its location and capacity profile, while `GET /mecs` allows the FRA to enumerate all registered MECs. For MEC-specific inspection, `GET /mecs/{mecId}` may be accessed by both the FRA and the corresponding MEC to retrieve detailed profile information, whereas `DELETE /mecs/{mecId}` is restricted to the FRA to explicitly deregister a MEC from federation participation.

The allocation lifecycle is orchestrated through the `/rounds` endpoints. `POST /rounds` is issued exclusively by the FRA to initialize a new allocation round with defined capacity and policy parameters. The status and metadata of allocation rounds can be monitored by both the FRA and MECs using `GET /rounds` and `GET /rounds/{roundId}`. The bidding phase is closed via `POST /rounds/{roundId}:lock`, which is strictly FRA-controlled, followed by `POST /rounds/{roundId}:run`, through which the FRA triggers execution of the MCKP-VCG solver.

Bid interactions are initiated by MECs through the `/bids` endpoints. `POST /bids` allows a MEC to submit tiered bids for a given round, while `PATCH /bids/{bidId}` enables partial bid updates prior to round locking. Bid withdrawal is supported through `DELETE /bids/{bidId}`. For transparency and verification, `GET /bids?roundId={id}` can be accessed by both the FRA and MECs to inspect all bids associated with a specific allocation round.

Allocation outcomes are exposed via the `/allocations` endpoints. `POST /allocations` is invoked by the FRA to publish the computed resource assignments and VCG-based payouts, while `GET /allocations/{allocationId}` allows both the FRA and MECs to retrieve detailed allocation and payment information. Final settlement is completed through `POST /allocations/{allocationId}:settle`, which is exclusively fired by the FRA to conclude the allocation cycle. Collectively, this role-constrained API design enforces a clear separation between control-plane authority and data-plane participation, ensuring that the *VERA* mechanism operates securely, transparently, and in alignment with CAMARA's federated interoperability model.

TABLE I
API ENDPOINTS AND THEIR FUNCTIONAL ROLES

Path	Verb	Purpose	Request	Response	Role
/mecs	POST	Register MEC	name, location, capacityProfile	mecId, status	FRA
/mecs	GET	List MECs	—	mecId, name, capacityProfile	FRA
/mecs/{mecId}	GET	Get MEC profile	—	mecId, profile	FRA/MEC
/mecs/{mecId}	DELETE	Deregister MEC	—	status	FRA/MEC
/rounds	POST	Create allocation round	capacity, policy, start, end	roundId, status	FRA
/rounds	GET	List rounds	—	roundId, status, policy	FRA/MEC
/rounds/{roundId}	GET	Round metadata	—	roundId, status, capacity	FRA/MEC
/rounds/{roundId}:lock	POST	Lock bidding	—	roundId, LOCKED	FRA
/rounds/{roundId}:run	POST	Run solver	—	opId	FRA
/bids	POST	Submit tiered bid	mecId, roundId, tiers	bidId, status	MEC
/bids?roundId={id}	GET	List bids	query: roundId	bidId, mecId, tiers	FRA/MEC
/bids/{bidId}	PATCH	Update bid	partial bid	bidId, status	MEC
/bids/{bidId}	DELETE	Withdraw bid	—	status	MEC
/allocations	POST	List allocations	—	allocationId, assignments, payouts	FRA/MEC
/allocations/{allocationId}	GET	Allocation details	—	assignment, payouts, totalValue	FRA/MEC
/allocations/{allocationId}:settle	POST	Settle allocation	—	allocationId, settled	FRA